



WM243-18 IM CW1

UniLearn Hub

Report

Information Management Coursework

Student: Syed Imad Uddin

August 11, 2026

Contents

1	Design and Export a Secure PostgreSQL Database Schema	2
1.1	ERD	2
1.2	Entities and relationships	2
1.3	Normalisation and constraint design	7
1.4	Roles, grants, and views	7
1.5	Verification	9
2	Configuration Folder for PostgreSQL Security	10
3	SQL Business Logic Executable Script	14
4	Testing of User Journeys	17
5	Security Design Decisions	17
6	AI Use Declaration	18

1 Design and Export a Secure PostgreSQL Database Schema

1.1 ERD

The diagram below is the Crow's Foot entity relationship diagram for the schema, covering all 16 tables and their foreign key relationships, including the zero or one cardinality on the nullable Notifications and Messages foreign keys.

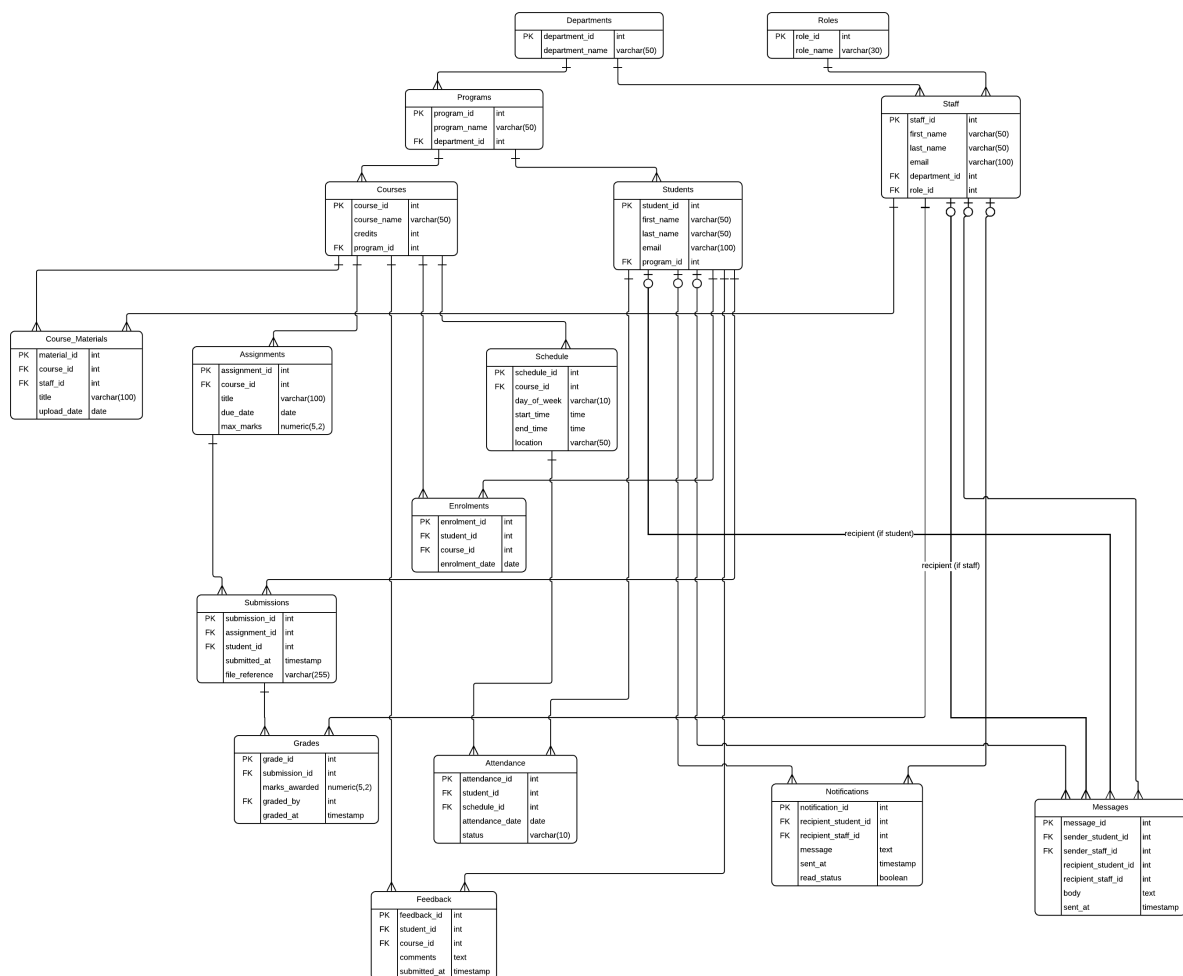


Figure 1: UniLearn Hub entity relationship diagram, Crow's Foot notation

1.2 Entities and relationships

I built the schema as 16 tables in a dedicated `unilearn` schema, not `public`, so the database can host other schemas alongside it without name collisions.

Four tables have no foreign keys and sit at the top of the dependency graph. `Departments` and `Roles` have no dependencies at all. `Programs` depends on `Departments`. `Staff` depends on `Departments` and `Roles`. Everything else builds on top of `Students` (depends on `Programs`) and `Courses` (depends on `Programs`).

I resolved the many to many relationships with two junction tables, Enrolments (student and course) and Submissions (assignment and student). Each has a UNIQUE constraint on the pair of foreign keys so the same relationship can't be inserted twice. Grades sits one level further down, one to one with Submissions through a UNIQUE NOT NULL foreign key, so a submission has at most one grade.

Notifications and Messages model a recipient, and for Messages a sender, that is either a student or a staff member, never both. I explain the constraint design behind this in Section 1.3.

Course_Materials, Assignments, Schedule, Attendance, and Feedback all hang off Courses or Students, recording the day to day activity the schema exists to track. I didn't store anything beyond what these features need. No scraped or inferred personal data, no column collecting more than the stated use case requires, in line with the brief's GDPR compliance expectation.

Every primary key is a plain INTEGER GENERATED ALWAYS AS IDENTITY rather than a UUID. This system has no need for multiple write nodes, so a sequential surrogate key keeps joins smaller and easier to read at this scale. Row creation order becomes visible in the key, which I judged acceptable here.

Table screenshots below (schema.sql), grouped by dependency layer.

```
CREATE SCHEMA unilearn;  
SET search_path TO unilearn;
```

Figure 2: Schema header

```
-- no dependencies  
  
CREATE TABLE Departments (  
    department_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,  
    department_name VARCHAR(50) UNIQUE NOT NULL  
);  
  
CREATE TABLE Roles (  
    role_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,  
    role_name VARCHAR(30) UNIQUE NOT NULL  
);
```

Figure 3: Departments, Roles

```
-- needs Departments / Roles

CREATE TABLE Programs (
  program_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
  program_name VARCHAR(50) UNIQUE NOT NULL,
  department_id INTEGER REFERENCES Departments(department_id) NOT NULL
);

CREATE TABLE Staff (
  staff_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
  first_name VARCHAR(50) NOT NULL,
  last_name VARCHAR(50) NOT NULL,
  email VARCHAR(100) UNIQUE NOT NULL,
  department_id INTEGER REFERENCES Departments(department_id) NOT NULL,
  role_id INTEGER REFERENCES Roles(role_id) NOT NULL
);
```

Figure 4: Programs, Staff

```
-- needs Programs

CREATE TABLE Students (
  student_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
  first_name VARCHAR(50) NOT NULL,
  last_name VARCHAR(50) NOT NULL,
  email VARCHAR(100) UNIQUE NOT NULL,
  program_id INTEGER REFERENCES Programs(program_id) NOT NULL
);

CREATE TABLE Courses (
  course_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
  course_name VARCHAR(50) NOT NULL,
  credits INTEGER NOT NULL,
  program_id INTEGER REFERENCES Programs(program_id) NOT NULL
);
```

Figure 5: Students, Courses

```
-- needs Courses / Staff

CREATE TABLE Course_Materials (
  material_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
  course_id INTEGER REFERENCES Courses(course_id) NOT NULL,
  staff_id INTEGER REFERENCES Staff(staff_id) NOT NULL,
  title VARCHAR(100) NOT NULL,
  upload_date DATE NOT NULL
);

CREATE TABLE Assignments (
  assignment_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
  course_id INTEGER REFERENCES Courses(course_id) NOT NULL,
  title VARCHAR(100) NOT NULL,
  due_date DATE NOT NULL,
  max_marks NUMERIC(5,2) NOT NULL CHECK (max_marks > 0)
);

CREATE TABLE Schedule (
  schedule_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
  course_id INTEGER REFERENCES Courses(course_id) NOT NULL,
  day_of_week VARCHAR(10) NOT NULL
  CHECK (day_of_week IN ('Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday', 'Sunday')),
  start_time TIME NOT NULL,
  end_time TIME NOT NULL,
  location VARCHAR(50) NOT NULL,
  CHECK (end_time > start_time)
);
```

Figure 6: Course_Materials, Assignments, Schedule

```
-- junction tables (many-to-many)

CREATE TABLE Enrolments (
  enrolment_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
  student_id INTEGER REFERENCES Students(student_id) NOT NULL,
  course_id INTEGER REFERENCES Courses(course_id) NOT NULL,
  enrolment_date DATE NOT NULL,
  UNIQUE (student_id, course_id)
);

CREATE TABLE Submissions (
  submission_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
  assignment_id INTEGER REFERENCES Assignments(assignment_id) NOT NULL,
  student_id INTEGER REFERENCES Students(student_id) NOT NULL,
  submitted_at TIMESTAMP NOT NULL,
  file_reference VARCHAR(255) NOT NULL,
  UNIQUE (assignment_id, student_id)
);
```

Figure 7: Enrolments, Submissions

```
-- needs Submissions / Staff

CREATE TABLE Grades (
  grade_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
  submission_id INTEGER REFERENCES Submissions(submission_id) UNIQUE NOT NULL,
  marks_awarded NUMERIC(5,2) NOT NULL CHECK (marks_awarded >= 0),
  graded_by INTEGER REFERENCES Staff(staff_id) NOT NULL,
  graded_at TIMESTAMP NOT NULL
);
```

Figure 8: Grades

```
-- needs Students / Schedule

CREATE TABLE Attendance (
  attendance_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
  student_id INTEGER REFERENCES Students(student_id) NOT NULL,
  schedule_id INTEGER REFERENCES Schedule(schedule_id) NOT NULL,
  attendance_date DATE NOT NULL,
  status VARCHAR(10) NOT NULL CHECK (status IN ('present','absent','late')),
  UNIQUE (student_id, schedule_id, attendance_date)
);

CREATE TABLE Feedback (
  feedback_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
  student_id INTEGER REFERENCES Students(student_id) NOT NULL,
  course_id INTEGER REFERENCES Courses(course_id) NOT NULL,
  comments TEXT NOT NULL,
  submitted_at TIMESTAMP NOT NULL
);
```

Figure 9: Attendance, Feedback

```
-- recipient/sender: student or staff -- nullable FKs + CHECK, not a type column

CREATE TABLE Notifications (
  notification_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
  recipient_student_id INTEGER REFERENCES Students(student_id),
  recipient_staff_id INTEGER REFERENCES Staff(staff_id),
  message TEXT NOT NULL,
  sent_at TIMESTAMP NOT NULL,
  read_status BOOLEAN NOT NULL DEFAULT FALSE,
  CHECK (
    (recipient_student_id IS NOT NULL AND recipient_staff_id IS NULL)
    OR
    (recipient_student_id IS NULL AND recipient_staff_id IS NOT NULL)
  )
);

CREATE TABLE Messages (
  message_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
  sender_student_id INTEGER REFERENCES Students(student_id),
  sender_staff_id INTEGER REFERENCES Staff(staff_id),
  recipient_student_id INTEGER REFERENCES Students(student_id),
  recipient_staff_id INTEGER REFERENCES Staff(staff_id),
  body TEXT NOT NULL,
  sent_at TIMESTAMP NOT NULL,
  CHECK (
    (sender_student_id IS NOT NULL AND sender_staff_id IS NULL)
    OR
    (sender_student_id IS NULL AND sender_staff_id IS NOT NULL)
  ),
  CHECK (
    (recipient_student_id IS NOT NULL AND recipient_staff_id IS NULL)
    OR
    (recipient_student_id IS NULL AND recipient_staff_id IS NOT NULL)
  )
);
```

Figure 10: Notifications, Messages

1.3 Normalisation and constraint design

The schema is in 3NF. Every non key column depends on the whole primary key and nothing but the key. There is no repeating group data, a course's schedule is its own table, not a set of day1, day2 columns, and no column is derivable from another column in the same row rather than stored directly.

Junction tables over multi valued columns. A student can enrol in many courses and a course has many students. Modelling this as an array column on either side would break 1NF and make listing every student in a course an unindexed scan. Enrolments and Submissions make both directions a plain indexed join, and their UNIQUE (a, b) constraints double as the business rule that a student can't enrol in the same course twice.

Nullable FK pair plus CHECK, not a subtype column. Notifications needed a recipient that is a student or a staff member. The alternative was a single recipient_type column plus a recipient_id with no foreign key at all, since one foreign key can't point at two different tables. That trades a real, database enforced foreign key for an application level convention with no constraint behind it. The CHECK on the two nullable FKs keeps referential integrity intact and rejects rows where both are null or both are set, at the database layer rather than the application layer. Messages repeats the pattern for both sender and recipient.

CHECK beyond NOT NULL. `Assignments.max_marks > 0`, `Grades.marks_awarded >= 0`, `Schedule.end_time > start_time`, and `Attendance.status IN (...)` all encode a rule that is true for every valid row, so I put it in the schema rather than reimplementing it in every client that writes to these tables.

Deliberately not built. `Grades.marks_awarded` has no CHECK against its parent assignment's `max_marks`. A CHECK constraint can only see columns within its own row, so comparing against a different table needs a trigger. I left this out to keep the submitted schema fully trigger free and easy to reason about.

1.4 Roles, grants, and views

I exported four least privilege roles alongside the tables. `admin_role` gets full access. `lecturer_role` gets SELECT, INSERT, UPDATE on the tables lecturers actively use, `Courses`, `Course_Materials`, `Assignments`, `Submissions`, `Grades`, `Feedback`, `Schedule`, `Attendance`, and SELECT only on reference data, `Students`, `Enrolments`, `Departments`, `Programs`. That gives read access to the context they need and write access only to what they own. `student_role` gets SELECT on course facing tables, SELECT, INSERT on `Feedback`, and no grant at all on `Grades`. `read_only_role` gets SELECT across the whole schema, for reporting and audit use.

I added two views to narrow what a role sees without exposing whole tables. `Student_Grade_View` joins `Submissions`, `Assignments`, and `Grades` but omits `graded_by`, so a student sees their mark without seeing which staff member assigned it. `Course_Roster_View` omits student email from the roster a lecturer sees. `student_role` has no grant on `Grades` itself, so the view is the only path to that data.

```
-- VIEWS restrict what a role can see without exposing whole tables

CREATE VIEW Student_Grade_View AS
SELECT
    su.student_id,
    a.assignment_id,
    a.title AS assignment_title,
    su.submitted_at,
    g.marks_awarded,
    a.max_marks,
    g.graded_at
FROM Submissions su
JOIN Assignments a ON a.assignment_id = su.assignment_id
LEFT JOIN Grades g ON g.submission_id = su.submission_id;
-- graded_by left out on purpose / a student doesn't need to see which
-- staff member graded them

CREATE VIEW Course_Roster_View AS
SELECT
    e.course_id,
    s.student_id,
    s.first_name,
    s.last_name,
    e.enrolment_date
FROM Enrolments e
JOIN Students s ON s.student_id = e.student_id;
```

Figure 11: `Student_Grade_View`, `Course_Roster_View`

```
-- ROLES, least privilege / USAGE needed before table grants work

CREATE ROLE admin_role LOGIN PASSWORD 'placeholder_change_me';
GRANT USAGE ON SCHEMA unilearn TO admin_role;
GRANT ALL PRIVILEGES ON ALL TABLES IN SCHEMA unilearn TO admin_role;

CREATE ROLE lecturer_role LOGIN PASSWORD 'placeholder_change_me';
GRANT USAGE ON SCHEMA unilearn TO lecturer_role;
GRANT SELECT, INSERT, UPDATE ON
    Courses, Course_Materials, Assignments, Submissions,
    Grades, Feedback, Schedule, Attendance
    TO lecturer_role;
GRANT SELECT ON Students, Enrolments, Departments, Programs TO lecturer_role;
GRANT SELECT ON Course_Roster_View TO lecturer_role;

CREATE ROLE student_role LOGIN PASSWORD 'placeholder_change_me';
GRANT USAGE ON SCHEMA unilearn TO student_role;
GRANT SELECT ON Courses, Schedule, Course_Materials, Programs TO student_role;
GRANT SELECT, INSERT ON Feedback TO student_role;
GRANT SELECT ON Submissions, Enrolments, Attendance TO student_role;
GRANT SELECT ON Student_Grade_View TO student_role;
-- Grades table is NOT granted / students see marks only through Student_Grade_View which hides graded_by
-- no RLS yet therefore grants read on ALL rows not just the student's own

CREATE ROLE read_only_role LOGIN PASSWORD 'placeholder_change_me';
GRANT USAGE ON SCHEMA unilearn TO read_only_role;
GRANT SELECT ON ALL TABLES IN SCHEMA unilearn TO read_only_role;
```

Figure 12: Roles and grants

1.5 Verification

I tested this against a live PostgreSQL 18 instance rather than assuming it from the schema text. `schema.sql` imported into a completely fresh database with zero errors, 16 CREATE TABLE, 2 CREATE VIEW, 4 CREATE ROLE plus their grants.

Table 1: Schema, constraint, privilege, and view verification

Test	Action	Result
FK rejection	Insert an Enrolments row referencing <code>student_id = 999</code> (doesn't exist)	Rejected. violates foreign key constraint
CHECK rejection	Insert a Notifications row with both recipient FKs NULL	Rejected. violates check constraint
CHECK rejection	Insert a Notifications row with both recipient FKs set	Rejected. Same constraint
CHECK acceptance	Insert a Notifications row with exactly one recipient FK set	Succeeds
UNIQUE rejection	Insert a duplicate (<code>student_id, course_id</code>) into Enrolments	Rejected. violates unique constraint
Privilege	<code>lecturer_role</code> . SELECT on Departments	Succeeds. Granted explicitly
Privilege	<code>lecturer_role</code> . INSERT/UPDATE on Departments	Rejected. permission denied
Privilege	<code>read_only_role</code> . INSERT on Departments	Rejected. permission denied
Privilege	<code>read_only_role</code> . SELECT on Departments	Succeeds
View security	<code>student_role</code> . SELECT on <code>Student_Grade_View</code>	Succeeds
View security	<code>student_role</code> . SELECT on Grades directly	Rejected. permission denied for table grades

One result is worth stating precisely. `lecturer_role` can read Departments, granted so a lecturer can see which department a course belongs to, but cannot write to it. Least privilege here means a role reads what it needs and writes only what it owns, not zero access outside its core tables.

2 Configuration Folder for PostgreSQL Security

`config/postgresql.conf` and `config/pg_hba.conf` are the two files that set the security posture. `server.crt` and `server.key` exist to support them, and `README.md` documents installation.

postgresql.conf. `listen_addresses = 'localhost'` means the server accepts connections only from this machine. I assumed the app tier runs alongside it rather than reaching it over a network interface. `port = 5432` and `max_connections = 100` are left at conventional values. `ssl = on` plus `ssl_cert_file/ssl_key_file` turns on TLS for any connection that isn't a local Unix socket. `password_encryption = scram-sha-256` selects the stronger of PostgreSQL's two password hash schemes over the legacy MD5 default. SCRAM never sends the password itself over the wire, even encrypted, only a challenge response proof. `logging_collector, log_connections, log_disconnections`, and `log_statement = 'mod'` together log every INSERT, UPDATE, DELETE, plus every connect and disconnect event, an audit trail a shared academic database needs. `autovacuum = on` keeps dead rows from updates and deletes from accumulating on disk. `shared_buffers` and `work_mem` are left at defaults appropriate for a small install.

pg_hba.conf. Rules are checked top to bottom, first match wins, so ordering carries as much meaning as the individual rules. `local` (Unix socket) connections need `scram-sha-256`. Even same machine access is authenticated, not trusted implicitly. Loopback TCP (`127.0.0.1/32, ::1/128`) must additionally use `hostssl`, not `host`. SSL is mandatory even for a connection that never leaves the machine, so encryption isn't something an administrator can skip by accident just by connecting locally. One illustrative subnet, `10.20.1.0/24`, is allowed to reach the `unilearn` database remotely, again `hostssl` only, standing in for a segmented app tier network. The last two lines reject every other address explicitly, `0.0.0.0/0` and `::/0`, rather than relying on the absence of a matching rule. The intent, that the database is not reachable from the open internet, is visible directly in the file rather than implied by default deny behaviour.

server.crt and server.key. A self signed certificate and key pair I generated for this coursework, providing the material `ssl = on` needs. I set `server.key` to `chmod 600`, readable only by its owner, since a private key readable by other local accounts is not really private.

Rule scope, not just rule presence. The remote rule is scoped to `hostssl unilearn all 10.20.1.0/24, not hostssl all all 10.20.1.0/24`. A host on that subnet still can't reach any database on the instance other than `unilearn`, which matters once more

than one project shares a server. The loopback rules stay all, since loopback is the trusted local admin path. No plaintext host rule exists anywhere in the file, including for loopback. TLS is enforced by there being no alternative, not by a setting that could be left disabled.

Installation, per config/README.md. Copy all four files into the PostgreSQL data directory, found with `SHOW data_directory;` or the `-D` path passed to `initdb`. Run `chmod 600 server.key`. Then `pg_ctl restart -D <data_directory>` for the settings to take effect. `postgresql.conf` changes need a restart rather than a reload because several of the settings here, `listen_addresses` and `ssl_cert_file`, are only read at server start.

`max_connections = 100` is a security relevant choice too, not just a capacity one. An unbounded limit turns a slow client or retry loop into a resource exhaustion path, since every connection holds server memory regardless of use. `log_connections` and `log_disconnections` exist for the same reason `log_statement = mod` does. An unusual pattern is only visible after the fact if it was logged at the time.

```

3
4 # Controls
5 #1. who can connect
6 #2. from where
7 #3. how
8
9
10 # Rules checked in order
11
12 # TYPE      DATABASE   USER        ADDRESS            METHOD
13
14 # this machine, local socket (app server running alongside the DB)
15 local      all        all         :::                  scram-sha-256
16
17 # A machine on the network (in this case the local machine bcs of 127... loopback TCP) must still authenticate w password
18 hostssl    all        all         127.0.0.1/32       scram-sha-256
19
20 hostssl    all        all         ::1/128             scram-sha-256
21
22 # app-tier subnet only, not the whole internal network
23 hostssl    unilearn   all         10.20.1.0/24       scram-sha-256
24
25 # explicitly reject everything else -- the database is not reachable
26 # from the open internet
27 host       all        all         0.0.0.0/0           reject
28 host       all        all         :::/0               reject

```

Figure 13: `pg_hba.conf`

```

4
5 # Connections
6 listen_addresses = 'localhost'
7 # only accept connections on this machine
8 # app server connects locally, not the internet
9
10 port = 5432
11 max_connections = 100
12
13 # SSL (encrypts data in transit)
14 ssl = on
15 ssl_cert_file = 'server.crt'
16 ssl_key_file = 'server.key'
17
18 # Authentication
19 password_encryption = scram-sha-256
20 # scram-sha-256 instead of the older md5 -- stronger password hashing
21
22 # Logging (who did what / when)
23 logging_collector = on
24 log_directory = 'log'
25 log_filename = 'postgresql-%Y-%m-%d.log'
26 log_connections = on
27 log_disconnections = on
28 log_statement = 'mod'
29
30 # 'mod' logs every INSERT/UPDATE/DELETE/etc
31 # this is what the database-level audit trail requires
32
33 # Memory / VACUUM (dead row cleanup)
34 shared_buffers = 128MB
35 work_mem = 4MB
36 autovacuum = on
37 # autovacuum removes dead rows left behind by updates/deletes
38 # it keeps the database from stale data staying on disk
39

```

Figure 14: postgresql.conf

```
FA > config > server.crt
-----BEGIN CERTIFICATE-----
MIDTDF3Cf2gAwIBAeIuL8rPb3JBL6Vn8ZoJmqPcY/0QogcwDQYJKoZIhvcNAQEL
BQAwGjEYMBYGA1UEAwPdW5pbGhvcmd4taHViLWlrMB4XDTI2MDgxMDI1MDI1Mlo=
DTI3MDgxMDI1MDI1MloGjEYMBYGA1UEAwPdW5pbGhvcmd4taHViLWlrMIBIIBJAN
BgkqhkiG9w0BAQEAQCAQ8AMIIBCGCAQEApg7+oxE8Nqb04pCfjYXcmgdlAQfA
SP2LwGId0IwDcFEypFp9/1hrFLkaJzgX/tfvpcdPLi2tZqlsQAYajZNTClefieQnA
h8ZQ/ma6hCEBeuurcGN8rJWkmukCB5W6Dgmudp7rB9yZCAFMQFbum5Mw8kWBAlI
oC35Pe2Y5wA1lWoppHcQnkZsE9zZgCaSB6DMMN29b3j6naSt5s8cFdZ2ugeVHWIrFb4
icyRiRUZxmcNcI0owZogfwAUZ7LlIKX+tiIwchGj/0tPTZNciDIgcHNjth7/LcWvVY
JeUuJA9db6QDf2Bbkbo0oCNHSVNBqdz3rCaH9U2yK9xwat6I9z2IHmvre33IDAQAB
o1MwUtdHq6BNVHQ4Ef9gU0Aw0J7c1mDL8S5wSGL4UpVw8wHwDVR0IjBggwFAUu
wGA0HIZ7c1mDL8S5wSGL4UpVw8wDwDVR0TAQH/BAUwAwEB/zANBgkqhkiG9w0B
AQEFAAOCAQEAZFspEo1JUkS1n9MpcvY7tEjYhL8Z20cEi.rxj2nughlf3LI5E6SCG
+Ba6TkiTHYNuScI9u0lqfPwkJ3wCp27YtDAM2QG3hNrR/v9PT87Y0yBII/fvhdI0
UPPYCW9HKJjXmnNxyAxEJ151pGFtnJyXPwvzOuMsraFFXJKKQE8g41YnfvadD6sL8
eAt3sRukz0n/LApkds1wJpRgHBU0I58/7vvwrN4tkvMUWzjDMhU8b5s8r5iHsa/
LRLudLUHdIO9jYEDwJL3wK81cWBIswYn/BGgaZ3q5akzqC9J1CnLaHE3c3GAtVZ
YCMUg/JJu37Yw0RtjokKcFvhQxcuDtXgIag==
-----END CERTIFICATE-----
```

Figure 15: server.crt

Verification. I couldn't safely apply this config to the live system PostgreSQL instance directly, it would need root and a full service restart of a shared instance. Instead I built an isolated throwaway PostgreSQL 18 instance in a scratch directory, dropped in the project's actual `postgresql.conf`, `pg_hba.conf`, `server.crt`, and

`server.key` unmodified, only the port number changed to avoid clashing with the running system instance, and tested it directly.

Table 2: Configuration verification against a live instance

Test	Result
Plaintext TCP connection (<code>sslmode=disable</code>)	Rejected. <code>pg_hba.conf</code> rejects connection ... no encryption
SSL TCP connection (<code>sslmode=require</code>)	Succeeds. TLSv1.3, TLS_AES_256_GCM_SHA384
Password hash scheme in use	SCRAM-SHA-256, confirmed via <code>pg_authid</code>
<code>log_statement = mod</code>	A CREATE TABLE, INSERT, DROP TABLE sequence appeared in the log file verbatim

3 SQL Business Logic Executable Script

`business_logic.sh` creates three PL/pgSQL functions against the `unilearn` schema, seeds minimal demonstration data, and calls each function to show its behaviour. `IM.sql` truncates every table with `RESTART IDENTITY CASCADE` without touching structure, so the two scripts together give a repeatable reset and demo cycle. This is what lets the executable be run more than once during marking.

I wrote the script as a `.sh` wrapping `psql` calls rather than a plain `.sql` file, per the brief's requirement that the executable be a shell script, not a SQL dump. Each stage, function creation, seeding, each demo call, is a separate `psql` invocation, so the output stays readable as labelled steps rather than one undifferentiated block of results.

`IM.sql` uses `TRUNCATE ... RESTART IDENTITY CASCADE` rather than `DELETE FROM` on each table. `TRUNCATE` is faster since it doesn't scan or log individual rows. `RESTART IDENTITY` resets every `IDENTITY` sequence, so a fresh run produces the same `student_id`, `enrolment_id`, and so on, as the last one. `CASCADE` clears tables in dependency order automatically, rather than me listing 16 tables by hand in an order that respects foreign keys.

Table 3: Business logic functions

Function	Behaviour	Why
<code>fn_enrol_student</code>	Enrols a student in a course. If already enrolled, raises a notice and returns the existing <code>enrolment_id</code> instead of erroring	Enrolling twice is a normal user action, a double click or a retry, not an error condition. I treated this as a business logic decision, not a schema one, so it lives in the function rather than relaxing the <code>UNIQUE</code> constraint
<code>fn_course_average</code>	Returns the average <code>marks_awarded</code> across a course's graded submissions, or <code>NULL</code> if nothing is graded yet	Returns <code>NULL</code> , not 0. An ungraded course is not the same as one averaging zero
<code>fn_record_attendance</code>	Inserts an attendance record. Does nothing on a duplicate, using <code>ON CONFLICT (student_id, schedule_id, attendance_date) DO NOTHING</code>	Marking the same session twice shouldn't crash a real register

A genuinely interesting result came from running `business_logic.sh` again against a database that still had earlier test data in it, rather than a freshly reset one. The

script's raw seed inserts failed loudly and correctly.

```
ERROR:  duplicate key value violates unique constraint "
        departments_department_name_key"
ERROR:  duplicate key value violates unique constraint "
        roles_role_name_key"
ERROR:  duplicate key value violates unique constraint "
        programs_program_name_key"
ERROR:  duplicate key value violates unique constraint "
        staff_email_key"
ERROR:  duplicate key value violates unique constraint "
        students_email_key"
```

But calling `fn_enrol_student(1, 1)` a second time against the same already seeded data didn't error. It printed `NOTICE: Student 1 already enrolled in course 1 (enrolment_id 1)` and returned the existing row. Same story for `fn_record_attendance`. Called twice for the same session, the second call silently did nothing, and `SELECT count(*) FROM Attendance` stayed at 1. This is the schema constraint layer and the business logic layer doing their jobs differently on purpose, not a bug. A raw duplicate insert should fail hard, while a function representing a real user action should handle an already done case gracefully. One note for anyone rerunning the demo without a reset in between, `Courses`, `Schedule`, and `Assignments` carry no `UNIQUE` constraint on their seed columns, so reseeding against them duplicates rows silently rather than erroring.

The script's own `set -e` didn't stop it partway through this, and that is expected, not a flaw. `set -e` aborts the bash script if a command it runs exits non zero, but each seeding block is a single `psql` invocation reading a whole heredoc of SQL statements. `psql` itself, without `ON_ERROR_STOP`, reports each failing statement and moves to the next one within that invocation, then still exits with success once the heredoc is done. That is why the five duplicate key errors printed as errors rather than killing the script, and why the run still finished and produced usable output for the rest of the demo.

I confirmed the full cycle was clean. `business_logic.sh`, then `IM.sql`, then `business_logic.sh` again, against a freshly reset database, produced identical output both times with zero errors, including the `fn_course_average` calls, which returned `NULL` before any grade existed and `42.00` immediately after I inserted one, in both runs.

None of the three functions is `SECURITY DEFINER`, so each runs with the privileges of whichever role calls it, not the privileges of whoever created it. I tested this directly. `student_role` calling `fn_enrol_student` for a pair it is already enrolled in works fine, since that path only selects. But `student_role` calling it for a genuinely new enrol-

ment fails, with `ERROR: permission denied for table enrolments`, because the function's insert runs as `student_role`, which only has `SELECT` on `Enrolments` (Section 1.4). As written, the enrolment function is only usable end to end by a role with write access to `Enrolments`, meaning `lecturer_role` or `admin_role`, not by the student it is named for. I am flagging this as a real, verified gap between the function's name and what `student_role` can actually do with it, rather than leaving it to look like it works for students just because it compiles and runs under a higher privileged role.

4 Testing of User Journeys

A student enrolls, is graded, and checks their result. `fn_enrol_student` creates the `Enrolments` row. A lecturer later inserts a `Submissions` row and a `Grades` row against it. The student then queries `Student_Grade_View` (Section 1.4) and sees their mark and the assignment's `max_marks`, but not `graded_by`. A direct `SELECT` on `Grades` from the same session is rejected (Section 1.5), which confirms the view is the only path, not a convention a direct connection could bypass.

A lecturer records attendance across repeated sessions. `fn_record_attendance` is called once per class. Calling it again for a session already marked does nothing rather than erroring (Section 3), so a lecturer resubmitting a register can't silently create duplicate rows or crash the request.

An unauthorised or unencrypted connection is refused before it can do anything. A plaintext TCP connection is rejected at the connection stage, before authentication runs (Section 2). A `read_only_role` session that does connect still cannot write (Section 1.5). Network and transport, then privilege, each have to pass independently for a write to happen.

5 Security Design Decisions

Least privilege roles (Section 1.4), TLS only remote access with an explicit deny all (Section 2), SCRAM SHA 256 hashing, and full write and connection audit logging form the core of the design, covered in Sections 1 and 2. View based column hiding (Section 1.4) keeps a sensitive column out of a role's reach without per column grants, which PostgreSQL doesn't support directly. I verified every access control decision here live under `SET ROLE`, not just by reading the grant statements.

Known limitations, not fixed, listed on purpose.

1. No Row Level Security. `student_role`'s grant on `Student_Grade_View` covers every row, not just that student's own. Enforcing an own rows only rule needs Postgres RLS policies, which I have not built yet.
2. No cross table `CHECK` on `Grades.marks_awarded` against `Assignments.max_marks`. This needs a trigger. I left it out to keep the submitted schema simple and fully trigger free.
3. Self signed SSL certificate. It encrypts the connection but isn't verifiable against a trusted CA the way a real deployment's certificate would be.
4. `CREATE ROLE ... PASSWORD 'placeholder_change_me'`. Every role password in schema

.sql is a placeholder, not a real credential.

5. 10.20.1.0/24 in pg_hba.conf is illustrative, standing in for a segmented app tier subnet. It is not a real deployed address range.

6 AI Use Declaration

I used AI for the following.

- Ran the schema, config, and business logic scripts against a live PostgreSQL instance to check my import, constraint, privilege, and SSL claims actually held up.
- Caught a factual error in my own privilege test notes, lecturer access to `Departments` is read only, not blocked outright as I had first written.
- Formatted the report into `LATEX`, matching the structure and style of an example report of mine.
- Helped me along with `business_logic.sh` and taught me bash as I went, since I had little experience with it. Also helped write the comments.
- Debugging help, mainly on the isolated PostgreSQL instance setup used for config verification.
- Grammar, punctuation, and phrasing edits throughout.
- Added the section cross references between related parts of the report.
- Mapped the schema design to the GDPR compliance point in the brief's notes.

Appendix: Full Code Listings

A.1 schema.sql

```
-- UniLearn Hub schema (draft)

CREATE SCHEMA unilearn;
SET search_path TO unilearn;

-- no dependencies

CREATE TABLE Departments (
    department_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY
        KEY,
    department_name VARCHAR(50) UNIQUE NOT NULL
);

CREATE TABLE Roles (
    role_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
    role_name VARCHAR(30) UNIQUE NOT NULL
);

-- needs Departments / Roles

CREATE TABLE Programs (
    program_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY
        KEY,
    program_name VARCHAR(50) UNIQUE NOT NULL,
    department_id INTEGER REFERENCES Departments(department_id
        ) NOT NULL
);

CREATE TABLE Staff (
    staff_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    email VARCHAR(100) UNIQUE NOT NULL,
    department_id INTEGER REFERENCES Departments(department_id
        ) NOT NULL,
    role_id INTEGER REFERENCES Roles(role_id) NOT NULL
);
```

```
-- needs Programs

CREATE TABLE Students (
    student_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY
        KEY,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    email VARCHAR(100) UNIQUE NOT NULL,
    program_id INTEGER REFERENCES Programs(program_id) NOT
        NULL
);

CREATE TABLE Courses (
    course_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY
        ,
    course_name VARCHAR(50) NOT NULL,
    credits INTEGER NOT NULL,
    program_id INTEGER REFERENCES Programs(program_id) NOT
        NULL
);

-- needs Courses / Staff

CREATE TABLE Course_Materials (
    material_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY
        KEY,
    course_id INTEGER REFERENCES Courses(course_id) NOT NULL,
    staff_id INTEGER REFERENCES Staff(staff_id) NOT NULL,
    title VARCHAR(100) NOT NULL,
    upload_date DATE NOT NULL
);

CREATE TABLE Assignments (
    assignment_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY
        KEY,
    course_id INTEGER REFERENCES Courses(course_id) NOT NULL,
    title VARCHAR(100) NOT NULL,
    due_date DATE NOT NULL,
    max_marks NUMERIC(5,2) NOT NULL CHECK (max_marks > 0)
);
```

```

CREATE TABLE Schedule (
    schedule_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY
        KEY,
    course_id INTEGER REFERENCES Courses(course_id) NOT NULL,
    day_of_week VARCHAR(10) NOT NULL
        CHECK (day_of_week IN ('Monday','Tuesday','Wednesday',
            'Thursday','Friday','Saturday','Sunday')),
    start_time TIME NOT NULL,
    end_time TIME NOT NULL,
    location VARCHAR(50) NOT NULL,
    CHECK (end_time > start_time)
);

-- junction tables (many-to-many)

CREATE TABLE Enrolments (
    enrolment_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY
        KEY,
    student_id INTEGER REFERENCES Students(student_id) NOT
        NULL,
    course_id INTEGER REFERENCES Courses(course_id) NOT NULL,
    enrolment_date DATE NOT NULL,
    UNIQUE (student_id, course_id)
);

CREATE TABLE Submissions (
    submission_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY
        KEY,
    assignment_id INTEGER REFERENCES Assignments(assignment_id
        ) NOT NULL,
    student_id INTEGER REFERENCES Students(student_id) NOT
        NULL,
    submitted_at TIMESTAMP NOT NULL,
    file_reference VARCHAR(255) NOT NULL,
    UNIQUE (assignment_id, student_id)
);

-- needs Submissions / Staff

CREATE TABLE Grades (
    grade_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,

```

```

        submission_id INTEGER REFERENCES Submissions(submission_id
        ) UNIQUE NOT NULL,
        marks_awarded NUMERIC(5,2) NOT NULL CHECK (marks_awarded
        >= 0),
        graded_by INTEGER REFERENCES Staff(staff_id) NOT NULL,
        graded_at TIMESTAMP NOT NULL
    );

-- needs Students / Schedule

CREATE TABLE Attendance (
    attendance_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY
    KEY,
    student_id INTEGER REFERENCES Students(student_id) NOT
    NULL,
    schedule_id INTEGER REFERENCES Schedule(schedule_id) NOT
    NULL,
    attendance_date DATE NOT NULL,
    status VARCHAR(10) NOT NULL CHECK (status IN ('present', '
    absent', 'late')),
    UNIQUE (student_id, schedule_id, attendance_date)
);

CREATE TABLE Feedback (
    feedback_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY
    KEY,
    student_id INTEGER REFERENCES Students(student_id) NOT
    NULL,
    course_id INTEGER REFERENCES Courses(course_id) NOT NULL,
    comments TEXT NOT NULL,
    submitted_at TIMESTAMP NOT NULL
);

-- recipient/sender: student or staff -- nullable FKs + CHECK,
-- not a type column

CREATE TABLE Notifications (
    notification_id INTEGER GENERATED ALWAYS AS IDENTITY
    PRIMARY KEY,
    recipient_student_id INTEGER REFERENCES Students(
    student_id),

```

```

    recipient_staff_id INTEGER REFERENCES Staff(staff_id),
    message TEXT NOT NULL,
    sent_at TIMESTAMP NOT NULL,
    read_status BOOLEAN NOT NULL DEFAULT FALSE,
    CHECK (
        (recipient_student_id IS NOT NULL AND
         recipient_staff_id IS NULL)
        OR
        (recipient_student_id IS NULL AND recipient_staff_id
         IS NOT NULL)
    )
);

CREATE TABLE Messages (
    message_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY
    KEY,
    sender_student_id INTEGER REFERENCES Students(student_id),
    sender_staff_id INTEGER REFERENCES Staff(staff_id),
    recipient_student_id INTEGER REFERENCES Students(
        student_id),
    recipient_staff_id INTEGER REFERENCES Staff(staff_id),
    body TEXT NOT NULL,
    sent_at TIMESTAMP NOT NULL,
    CHECK (
        (sender_student_id IS NOT NULL AND sender_staff_id IS
         NULL)
        OR
        (sender_student_id IS NULL AND sender_staff_id IS NOT
         NULL)
    ),
    CHECK (
        (recipient_student_id IS NOT NULL AND
         recipient_staff_id IS NULL)
        OR
        (recipient_student_id IS NULL AND recipient_staff_id
         IS NOT NULL)
    )
);

-- VIEWS -- restrict what a role can see without exposing
-- whole tables

```

```
CREATE VIEW Student_Grade_View AS
SELECT
    su.student_id,
    a.assignment_id,
    a.title AS assignment_title,
    su.submitted_at,
    g.marks_awarded,
    a.max_marks,
    g.graded_at
FROM Submissions su
JOIN Assignments a ON a.assignment_id = su.assignment_id
LEFT JOIN Grades g ON g.submission_id = su.submission_id;
-- graded_by left out on purpose -- a student doesn't need to
-- see which
-- staff member graded them

CREATE VIEW Course_Roster_View AS
SELECT
    e.course_id,
    s.student_id,
    s.first_name,
    s.last_name,
    e.enrolment_date
FROM Enrolments e
JOIN Students s ON s.student_id = e.student_id;
-- student email left out -- roster doesn't need it

-- ROLES, least privilege -- USAGE needed before table grants
work

CREATE ROLE admin_role LOGIN PASSWORD 'placeholder_change_me';
GRANT USAGE ON SCHEMA unilearn TO admin_role;
GRANT ALL PRIVILEGES ON ALL TABLES IN SCHEMA unilearn TO
    admin_role;

CREATE ROLE lecturer_role LOGIN PASSWORD '
    placeholder_change_me';
GRANT USAGE ON SCHEMA unilearn TO lecturer_role;
GRANT SELECT, INSERT, UPDATE ON
    Courses, Course_Materials, Assignments, Submissions,
```

```

    Grades, Feedback, Schedule, Attendance
    TO lecturer_role;
GRANT SELECT ON Students, Enrolments, Departments, Programs TO
    lecturer_role;
GRANT SELECT ON Course_Roster_View TO lecturer_role;

CREATE ROLE student_role LOGIN PASSWORD 'placeholder_change_me
    ';
GRANT USAGE ON SCHEMA unilearn TO student_role;
GRANT SELECT ON Courses, Schedule, Course_Materials, Programs
    TO student_role;
GRANT SELECT, INSERT ON Feedback TO student_role;
GRANT SELECT ON Submissions, Enrolments, Attendance TO
    student_role;
GRANT SELECT ON Student_Grade_View TO student_role;
-- Grades table itself is NOT granted -- students see marks
-- only through
-- Student_Grade_View, which hides graded_by
-- no RLS yet -- this grants read on ALL rows, not just the
-- student's own

CREATE ROLE read_only_role LOGIN PASSWORD '
    placeholder_change_me';
GRANT USAGE ON SCHEMA unilearn TO read_only_role;
GRANT SELECT ON ALL TABLES IN SCHEMA unilearn TO
    read_only_role;

```

A.2 business_logic.sh

```

#!/bin/bash
# Creates the business logic functions, seeds demo data, runs
# each function.
# Usage: ./business_logic.sh [database_name]. Run IM.sql
# between runs to reset.

set -e

DB_NAME="${1:-unilearn_hub}"
PSQL="psql -d $DB_NAME"

echo "=== Creating functions ==="

```

```

$PSQL <<'SQL'
SET search_path TO unilearn;

-- Enrols a student. Already enrolled -> notice + existing id,
  not an error.
CREATE OR REPLACE FUNCTION fn_enrol_student(p_student_id INT,
  p_course_id INT)
RETURNS INT AS $$
DECLARE
  v_enrolment_id INT;
BEGIN
  SELECT enrolment_id INTO v_enrolment_id
  FROM Enrolments
  WHERE student_id = p_student_id AND course_id =
    p_course_id;

  IF v_enrolment_id IS NOT NULL THEN
    RAISE NOTICE 'Student % already enrolled in course % (
      enrolment_id %)',
      p_student_id, p_course_id, v_enrolment_id;
    RETURN v_enrolment_id;
  END IF;

  INSERT INTO Enrolments (student_id, course_id,
    enrolment_date)
  VALUES (p_student_id, p_course_id, CURRENT_DATE)
  RETURNING enrolment_id INTO v_enrolment_id;

  RETURN v_enrolment_id;
END;
$$ LANGUAGE plpgsql;

-- Average mark for a course. NULL if nothing graded yet, not
  0.
CREATE OR REPLACE FUNCTION fn_course_average(p_course_id INT)
RETURNS NUMERIC AS $$
DECLARE
  v_average NUMERIC;
BEGIN
  SELECT AVG(g.marks_awarded) INTO v_average

```

```

        FROM Grades g
        JOIN Submissions su ON su.submission_id = g.submission_id
        JOIN Assignments a ON a.assignment_id = su.assignment_id
        WHERE a.course_id = p_course_id;

        RETURN v_average;
END;
$$ LANGUAGE plpgsql;

-- Records attendance. Duplicate session -> no-op, not an
error.
CREATE OR REPLACE FUNCTION fn_record_attendance(
    p_student_id INT, p_schedule_id INT, p_date DATE, p_status
        VARCHAR
)
RETURNS VOID AS $$
BEGIN
    INSERT INTO Attendance (student_id, schedule_id,
        attendance_date, status)
    VALUES (p_student_id, p_schedule_id, p_date, p_status)
    ON CONFLICT (student_id, schedule_id, attendance_date) DO
        NOTHING;
END;
$$ LANGUAGE plpgsql;
SQL

echo "=== Seeding demo data ==="

$PSQL <<'SQL'
SET search_path TO unilearn;

INSERT INTO Departments (department_name) VALUES ('Computer
    Science');
INSERT INTO Roles (role_name) VALUES ('Lecturer');
INSERT INTO Programs (program_name, department_id) VALUES ('
    BSc Computing', 1);
INSERT INTO Staff (first_name, last_name, email, department_id
    , role_id)
    VALUES ('Beverly', 'Crusher', 'bcrusher@test.edu', 1, 1);
INSERT INTO Students (first_name, last_name, email, program_id
    )

```

```

VALUES ('Jean', 'Picard', 'jpicard@test.edu', 1);
INSERT INTO Courses (course_name, credits, program_id) VALUES
('Databases', 15, 1);
INSERT INTO Schedule (course_id, day_of_week, start_time,
end_time, location)
VALUES (1, 'Monday', '09:00', '11:00', 'Room A1');
INSERT INTO Assignments (course_id, title, due_date, max_marks
)
VALUES (1, 'Assignment 1', '2026-09-01', 50);
SQL

echo "=== fn_enrol_student ==="
$PSQL -c "SET search_path TO unilearn; SELECT fn_enrol_student
(1, 1) AS enrolment_id;"
echo "--- calling again: notice, not a duplicate row ---"
$PSQL -c "SET search_path TO unilearn; SELECT fn_enrol_student
(1, 1) AS enrolment_id;"

echo "=== fn_course_average, before grading -> NULL ==="
$PSQL -c "SET search_path TO unilearn; SELECT
fn_course_average(1) AS average_before_grading;"

$PSQL <<'SQL'
SET search_path TO unilearn;
INSERT INTO Submissions (assignment_id, student_id,
submitted_at, file_reference)
VALUES (1, 1, now(), '/uploads/a1.pdf');
INSERT INTO Grades (submission_id, marks_awarded, graded_by,
graded_at)
VALUES (1, 42, 1, now());
SQL

echo "=== fn_course_average, after grading -> 42.00 ==="
$PSQL -c "SET search_path TO unilearn; SELECT
fn_course_average(1) AS average_after_grading;"

echo "=== fn_record_attendance ==="
$PSQL -c "SET search_path TO unilearn; SELECT
fn_record_attendance(1, 1, CURRENT_DATE, 'present');"
$PSQL -c "SET search_path TO unilearn; SELECT * FROM
Attendance;"

```

```
echo "--- recording the same session again: no error, no
duplicate ---"
$PSQL -c "SET search_path TO unilearn; SELECT
    fn_record_attendance(1, 1, CURRENT_DATE, 'present');"
$PSQL -c "SET search_path TO unilearn; SELECT count(*) AS
    attendance_row_count FROM Attendance;"

echo "=== Done ==="
```

A.3 IM.sql

```
-- IM.sql -- resets all UniLearn Hub data between test runs.
-- Structure (tables/views/roles/functions) is untouched --
-- only rows
-- are cleared, so the business logic script can be re-run
-- repeatedly
-- against a clean state.

SET search_path TO unilearn;

TRUNCATE TABLE
    Departments, Roles, Programs, Staff, Students, Courses,
    Course_Materials, Assignments, Schedule, Enrolments,
    Submissions,
    Grades, Attendance, Feedback, Notifications, Messages
RESTART IDENTITY CASCADE;
```

A.4 config/postgresql.conf

```
# UniLearn Hub
# Security-relevant settings only
# Anything not listed here uses PostgreSQL's default settings

# Connections
listen_addresses = 'localhost'
# only accept connections on this machine
# app server connects locally, not the internet

port = 5432
max_connections = 100
```

```
# SSL (encrypts data in transit)
ssl = on
ssl_cert_file = 'server.crt'
ssl_key_file = 'server.key'

# Authentication
password_encryption = scram-sha-256
# scram-sha-256 instead of the older md5 -- stronger password
  hashing

# Logging (who did what / when)
logging_collector = on
log_directory = 'log'
log_filename = 'postgresql-%Y-%m-%d.log'
log_connections = on
log_disconnections = on
log_statement = 'mod'

# 'mod' logs every INSERT/UPDATE/DELETE/etc
# this is what the database-level audit trail requires

# Memory / VACUUM (dead row cleanup)
shared_buffers = 128MB
work_mem = 4MB
autovacuum = on
# autovacuum removes dead rows left behind by updates/deletes
# it keeps the database from stale data staying on disk
```

A.5 config/pg_hba.conf

```
# UniLearn Hub

# Controls
#1. who can connect
#2. from where
#3. how

# Rules checked in order
```

```

# TYPE      DATABASE    USER      ADDRESS      METHOD

# this machine, local socket (app server running alongside the
# DB)
local       all          all        all           scram-sha-256

# A machine on the network (in this case the local machine
# bcs of 127... loopback TCP) must still authenticate w
# password
hostssl     all          all        127.0.0.1/32  scram-sha-256
hostssl     all          all        ::1/128       scram-sha-256

# app-tier subnet only, not the whole internal network
hostssl     unilearn    all        10.20.1.0/24  scram-sha-256

# explicitly reject everything else -- the database is not
# reachable
# from the open internet
host        all          all        0.0.0.0/0     reject
host        all          all        ::/0          reject

```

A.6 config/README.md

```

# UniLearn Hub -- config

PostgreSQL security configuration for the UniLearn Hub
database.

## Files

**postgresql.conf**
Server-wide settings. Restricts network exposure ('
listen_addresses'),
turns on SSL, sets stronger password hashing ('scram-sha-256')
,
enables full write logging ('log_statement = mod') for audit
purposes,
and keeps autovacuum on so dead rows from updates/deletes don't
linger on disk.

```

```
**pg_hba.conf**  
Controls who can connect, from where, and how. Local/loopback  
connections must authenticate with 'scram-sha-256'. Only a  
specific  
trusted subnet is allowed to connect remotely, and only over  
SSL  
( 'hostssl', not 'host' ). Everything else is explicitly  
rejected --  
the database is not reachable from the open internet.
```

```
**server.key / server.crt**  
Self-signed key/cert pair, backs the SSL that 'postgresql.conf  
' and  
'pg_hba.conf' require. 'server.key' is in the coursework  
submission  
zip but left out of this public GitHub repo, since a real  
private key  
has no business being published even when it's a coursework  
placeholder.
```

Installation

1. Copy all four files into the PostgreSQL data directory (
found via
'SHOW data_directory;' on an existing install, or the '-D'
path
passed to 'initdb').
2. Set correct permissions on the private key:
'chmod 600 server.key'
3. Restart PostgreSQL for the settings to take effect:
'pg_ctl restart -D <data_directory>'

Notes

- '10.20.1.0/24' in 'pg_hba.conf' is illustrative -- stands in
for a
segmented app-tier subnet, not a real deployed range.
Replace before
deploying.

- Self-signed cert: encrypts, but isn't verifiable like a CA-signed one.
- 'CREATE ROLE' passwords in 'schema.sql' are placeholders, not real credentials.
- Tested against a live PostgreSQL 18 instance: SSL, 'scram-sha-256', and 'log_statement=mod' all confirmed working; plaintext (non-SSL) connections correctly rejected by 'pg_hba.conf'.